

DIRECTED ENERGY ENGAGEMENT DIGITAL TWIN

AEROSPACE MATHEMATICS APPENDIX

Implementation-aligned governing equations, reduced-order models, numerical methods, state estimation, thermal/power physics, and V&V; structure from app.py v3.

IMPLEMENTATION SCOPE

Domains: rigid-body dynamics, FRD $\leftrightarrow$ FLU conversion, atmosphere, beam geometry, pointing, target thermal response, platform power/thermal response, component thermal network, 1-D Fourier conduction, EKF, Monte Carlo UQ, and requirements evidence.

Canonical frame: FLU, x-forward, y-left, z-up.

Effect metric: Estimated Thermal Effect Index, not probability.

CREDIBILITY BOUNDARY

Physics-informed multidisciplinary digital-engineering demonstrator. It does not claim operational weapon validation, high-fidelity missile aerodynamics, wave-optics propagation, lethality prediction, or hardware-qualified thermal performance.

NASA GTM: model-to-model reference benchmarking.

NOAA IGRA: atmospheric-state observational comparison only.

CLOSED-LOOP FUNCTIONAL CONTEXT

Detect $\rightarrow$ Identify $\rightarrow$ Track $\rightarrow$ Decide $\rightarrow$ Point / Aim $\rightarrow$ Deliver Energy $\rightarrow$ Assess Effect $\rightarrow$ Re-engage

The mathematics below supports this loop through coupled dynamics, sensing, estimation, atmosphere, beam, thermal, power, uncertainty, and evidence models.

MODEL • SIMULATE • VERIFY • BENCHMARK • DECIDE

1. Reference Frames and Rigid-Body Dynamics

RIGID-BODY CORE

$$\begin{aligned}\dot{\mathbf{r}}_i &= \mathbf{C}_{\beta i} \mathbf{V}_\beta \\ \mathbf{V}_\beta &= \mathbf{F}_\beta / m + \mathbf{C}_{i\beta} \mathbf{g}_i - \boldsymbol{\omega}_\beta \times \mathbf{V}_\beta \\ \mathbf{I} \dot{\boldsymbol{\omega}}_\beta &= \mathbf{M}_\beta - \boldsymbol{\omega}_\beta \times (\mathbf{I} \boldsymbol{\omega}_\beta)\end{aligned}$$

$\mathbf{r}_i$: inertial position; $\mathbf{V}_\beta$: body velocity; $\mathbf{F}_\beta$: body force; $\mathbf{M}_\beta$: body moment; $\mathbf{I}$: inertia tensor; $\boldsymbol{\omega}_\beta$: body angular rate.

NASA/GTM FRD → APP FLU

$$\begin{aligned}\mathbf{S} &= \text{diag}(1, -1, -1) \\ \mathbf{v}_{\text{FLU}} &= \mathbf{S} \mathbf{v}_{\text{FRD}} \\ \mathbf{M}_{\text{FLU}} &= \mathbf{S} \mathbf{M}_{\text{FRD}} \\ \mathbf{C}_{\text{FLU}} &= \mathbf{S} \mathbf{C}_{\text{FRD}} \mathbf{S}^T\end{aligned}$$

FRD: x forward, y right, z down. FLU: x forward, y left, z up. Attitude is transformed as a basis change; quaternions remain authoritative internally.

GENERIC AERODYNAMIC LOAD

$$\mathbf{F}_{\text{drag}} = -\frac{1}{2} \rho V^2 C_D A \hat{\mathbf{v}}$$

Generic low-order drag only. Matched NASA GTM reference loads are treated as resultant non-gravitational body loads and are not augmented with this generic drag term.

MODEL LIMIT

The generic 6-DOF implementation is a low-order rigid-body dynamics model. It is not presented as a validated aircraft or missile aerodynamic model.

2. Atmosphere, Pointing, and Beam Geometry

BEER-LAMBERT EXTINCTION

$$\begin{aligned}\beta_{\text{aer},550} &= 3.912 / V_{\text{vis}} \\ \beta_{\text{aer}} &= \beta_{\text{aer},550} (0.55/\lambda)^{\text{\AA}} \\ \beta_{\text{R}} &= 0.0116 (0.55/\lambda)^4 \\ \beta_{\text{H}} &= \beta_{\text{H},100} \cdot \text{clamp}(\text{RH}/100) \\ \tau &= \exp[-(\beta_{\text{aer}} + \beta_{\text{R}} + \beta_{\text{H}})R]\end{aligned}$$

Coefficients use km^{-1} and R uses km . This is phenomenological reduced-order extinction, not MODTRAN or line-by-line radiative transfer.

REDUCED-ORDER BEAM FOOTPRINT

$$\begin{aligned}\theta_{\text{diff}} &= M^2 \lambda / (\pi w_0) \\ \theta_{\text{eff}} &= \sqrt{(\theta_{\text{diff}})^2 + \theta_{\text{add}}^2 + \theta_{\text{turb}}^2} \\ r_{\text{spot}} &= \sqrt{[w_0^2 + (\theta_{\text{eff}} R)^2 + (\theta_{\text{point}} R)^2]} \\ I_{\text{avg}} &= P_{\text{target}} / (\pi r_{\text{spot}}^2) \\ P_{\text{target}} &= P_{\text{opt,actual}} \cdot \eta_{\text{optics}} \cdot \tau\end{aligned}$$

Pointing uncertainty expands the footprint; it is not an explicit multiplicative pointing-efficiency term.

POINTING + SERVO

$$\begin{aligned}\sigma_{\text{point}} &= \sqrt{(\sigma_{\text{base}}^2 + \sigma_{\text{wind}}^2 + \sigma_{\text{turb}}^2 + \sigma_{\text{track}}^2)} \\ e_{\text{eff}} &= \sqrt{(\sigma_{\text{point}}^2 + e_{\text{servo}}^2)} \\ \Delta\theta_{\text{des}} &= e[1 - \exp(-\Delta t/\tau_{\text{servo}})] \\ |\Delta\theta| &\leq \omega_{\text{max}} \Delta t\end{aligned}$$

Stochastic RSS contributors and deterministic beam-director lag are kept conceptually separate.

NOAA IGRA constrains atmospheric-state variables only; it does not validate optical transmission.

3. Target Thermal Response, Effect Metric, and Power

PRIMARY TARGET THERMAL MODEL

$$C_A \, dT/dt = \alpha \, I_{\text{avg}} - h_A(T - T_{\text{amb}})$$

$$\tau_{\text{th}} = C_A / h_A$$

$$\Delta T_{\text{ss}} = \alpha \, I_{\text{avg}} / h_A$$

C_A [kJ/m²-K]; I_{avg} [kW/m²]; h_A [kW/m²-K].

$$I_{\text{thermal}} = \text{clamp}(\Delta T / \Delta T_{\text{fail,eff}}, 0, 1)$$

Estimated Thermal Effect Index is a normalized engineering index, not P(effect), P(kill), or lethality probability.

GENERATOR + STORAGE

$$P_{\text{store,avail}} = \min(P_{\text{store,max}}, 3600E_{\text{store}}/t_{\text{dwell}})$$

$$P_{\text{elec,actual}} = \min(P_{\text{elec,req}}, P_{\text{gen}} + P_{\text{store,avail}})$$

$$P_{\text{opt,actual}} = \eta_{\text{WP}} P_{\text{elec,actual}}$$

$$E_{\text{used}} = P_{\text{store,draw}} t_{\text{dwell}} / 3600$$

E in kWh, t in s, P in kW. The 3600 conversion is required for unit consistency.

PLATFORM HEAT BALANCE

$$Q_{\text{conv}} = P_{\text{elec,actual}} - P_{\text{opt,actual}}$$

$$Q_{\text{optics}} = P_{\text{opt,actual}}(1 - \eta_{\text{optics}})$$

$$Q_{\text{internal}} = Q_{\text{conv}} + Q_{\text{optics}}$$

$$\eta_{\text{cool}} = \text{clamp}[(T_{\text{coolant}} - T_{\text{amb}})/(T_{\text{limit}} - T_{\text{amb}})]$$

$$Q_{\text{cool}} = Q_{\text{cool,max}} \eta_{\text{cool}}$$

Implemented ambient-limited lumped cooling abstraction.

AIMPOINT MARGIN

$$\theta_{\text{allow}} \approx r_{\text{char}}/R$$

$$m_{\text{aim}} = \text{clamp}\{\exp[-\frac{1}{2}(e_{\text{eff}}/\theta_{\text{allow}})^2]\}$$

Dimensionless angular margin index. It is not a probability.

4. Component Thermal Network and 1-D Fourier Solver

MULTI-NODE COMPONENT NETWORK

$$C_i dT_i/dt = Q_i - \sum_j G_{ij}(T_i - T_j) - Q_{\text{sink},i}$$

Implemented units: C_i in kJ/K; G_{ij} in kW/K; Q in kW. Nodes: Laser Source, Beam Director / Optics, Power Electronics, Energy Store, Coolant Loop, Heat Exchanger.

HX sink scales with positive temperature head above ambient and is capped by commanded cooling capacity. Energy Store receives a generic $0.015 \times \text{storage-draw heat proxy}$.

FOURIER CONDUCTION

$$\rho c_p \partial T / \partial t = \partial / \partial x (k \partial T / \partial x)$$

$$\alpha = k / (\rho c_p)$$

$$\text{Front: } -k \partial T / \partial x = q''_{\text{abs}} - h_f(T_s - T_\infty)$$

$$\text{Back: } -k \partial T / \partial x = -h_b(T_b - T_{\text{fluid}})$$

Positive absorbed heat flux is defined into the slab.

EXPLICIT STABILITY

$$Fo = \alpha \Delta t / \Delta x^2 \leq 0.45$$

$$\Delta t [\alpha / \Delta x^2 + h / (\rho c_p \Delta x)] \leq 0.45$$

Internal timestep is limited by interior and front/back boundary criteria. The solver steps exactly to output times and the duration boundary.

Readiness: HOLD if a component reaches limit; CAUTION if minimum peak margin < 5 °C; READY otherwise; INVALID if initially at/above limit. No crossing within the horizon is reported as right-censored, e.g. >120 s.

5. State Estimation, UQ, and Engineering Evidence

EXTENDED KALMAN FILTER

$$\begin{aligned}\hat{\mathbf{x}}_k|_{k-1} &= f(\hat{\mathbf{x}}_{k-1}|_{k-1}, \mathbf{u}_{k-1}) \\ \mathbf{P}_k|_{k-1} &= \mathbf{F}_k \mathbf{P}_{k-1}|_{k-1} \mathbf{F}_k^T + \mathbf{Q}_k \\ \mathbf{K}_k &= \mathbf{P}^- \mathbf{H}^T (\mathbf{H} \mathbf{P}^- \mathbf{H}^T + \mathbf{R})^{-1} \\ \hat{\mathbf{x}}^+ &= \hat{\mathbf{x}}^- + \mathbf{K} [\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}^-)] \\ \mathbf{P}^+ &= (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}^- (\mathbf{I} - \mathbf{K} \mathbf{H})^T + \mathbf{K} \mathbf{R} \mathbf{K}^T\end{aligned}$$

Joseph stabilized covariance update; covariance is symmetrized after update. NIS is computed from innovation and innovation covariance.

MONTE CARLO / UQ

$$\mathbf{X} = g(\theta, \xi) \\ \{\mathbf{X}^1, \mathbf{X}^2, \dots, \mathbf{X}^N\} \rightarrow \text{empirical output distribution}$$

Monte Carlo propagates modeled uncertainty through the coupled simulation. Output distributions are uncertainty-analysis results, not automatically calibrated real-world probabilities.

Higher credibility requires documented distributions, provenance, correlations, and epistemic vs. aleatory classification.

TRACEABILITY BASELINE

BL-001 | Version 1.0 | Draft engineering baseline

Need → System Requirement → Allocated Function/Model → Verification

Method/Case → Evidence → Metric → Threshold → Status.

The application resolves textual trace IDs against defined needs and system-requirement registries.

V&V; SEMANTICS

Verification: analytical checks, stability, invariants, regression, requirement evaluation.

NASA GTM: model-to-model reference benchmarking.

NOAA IGRA: atmospheric-state observational comparison.

Validation: reserved for evidence supporting intended-use adequacy; not inferred from the two comparisons above.

Source basis: Directed_Energy_Engagement_Digital_Twin_app_v3.py. This appendix follows the implementation rather than the earlier promotional infographic.